A-A-54251
23 November 1990

COMMERCIAL ITEM DESCRIPTION

LIGHT SURGICAL FIELD

The General Services Administration has authorized the use of this commercial item description

This Commercial Item Description covers a portable, floor-standing (with casters), operating and examining light

Salient Characteristics:

The surgical field light shall be a portable, floor-standing (with casters), operating and examining light equipped for battery operation to provide continued light output in areas with intermittent input power

It shall be suitable for use within hardwall shelters in military field medical facilities as well as all fixed medical and dental facilities M U S T units and Forward Aid Stations

This unit shall provide surgical quality light for use in applications where high intensities are needed, but field size requirements are less critical than major surgery

It shall be designed to assemble and disassemble in five minutes or less for operation and storage within its own reusable shipping container

Design and construction

The design shall permit the lighthead to be raised to 6'6" and lowered to 3'10" above the finished floor The extension arm shall be 26" from the mounting post to the lighthead center The extension arm shall pivot 90° around the mounting tube The lamphead yoke shall rotate 180° around the extension arm The lamphead shall rotate 180° within the yoke

The unit shall weigh not more than 60 lbs without batteries

AMSC N/A FSC 6530

DISTRIBUTION STATEMENT A Approved for public release distribution is unlimited

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and 95 lbs with batteries.

Base:

The base of the light shall contain the electronics and batteries. It shall be constructed of steel and have no openings on the bottom surface or sides for depth of 3 inches. The light shall rest evenly on a flat surface without rocking. It shall not have any burrs, slivers, or sharp edges. The base shall be so configured, and sufficiently heavy, that the light shall not tip over throughout the full range of lighthead and extension arm movement. The base shall have as a minimum two casters, a minimum of 10" in diameter, to provide for easy portability and transport over uneven floor surfaces of field hospitals or aid stations.

Upright mounting tube.

The upright mounting tube shall be 1-1/2" diameter tubing with a wall thickness of 1/8". The tube shall be no more than 36" long. The tube shall be designed with a pinned, quick lock connector at the point of attachment to the base and on the tube interconnection point.

Extension arm.

The extension arm shall be made of extruded 1" x 2" aluminum and shall be of conventional design. The arm shall be attached to the upper end of the upright mounting tube.

Utility requirements.

The light shall operate within a range of 95 through 135V, 2.5 AMP and 190 through 270V, 1.25 AMP, with not more than a 5 percent change in footcandle intensity, when plugged into a standard three-prong grounded receptacle. Input voltage range shall be determined and selected automatically by controls within the light fixture. The light shall be equipped with an automatic switching device for battery operation in the event of an interruption of input power. The light shall operate on 50 or 60 hertz power systems.

For battery operation, two 12 volt, deep cycle, sealed batteries are required to achieve a nominal 24V lamp voltage. The batteries shall supply 2.50 AMPS at 24 volts for a minimum

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of four (4) hours. The cut-off voltage for testing shall be 87.5 percent of the nominal voltage (24V) or 21 volts. The battery charger circuit shall be incorporated into the solid state controls (21V to fully charge in 8 hours maximum).

The light shall be supplied with a flexible, 3 conductor rubber covered, No 18 American wire gauge or heavier, power cord, 11 feet in length; measured from plug to base The three-prong hospital grade plug shall be as specified in UL 498, which is compatible with the receptacle specified in ANSI C73 11. Suitable strain relief shall be provided in accordance with UL 544

Lighthead

The lamphead shall be equipped with a 60 watt, 24 volt quartz halogen lamp having a rated average life of 500 hours It shall be capable of providing approximately 3500 footcandles of illumination at the center of the spot 36" from the lower rim of the lamphead when lighthead is in a horizontal position Color correction shall be approximately 4000° kelvin The irradiance factor shall not exceed 5.0 micro-watts per square centimeter per footcandle The light shall have a universal focus design so the beam will be in proper focus through the normal range of use The lighthead shall be 14" across and approximately square in shape The lens area shall be approximately 103 square inches, circular in shape The lamphead shall be 7 inches high and designed so that it is shatterproof to the point that no particles are emitted from the lamphead should a bulb self destruct.

The lighthead shall be made of polycarbonated resins The reflector shall be dichroic coated polyetherimide, 10 5 inches diameter, which transmits lamp heat upward and reflects cool visible light onto the surgical field The lighthead shall have a closed dome construction sufficiently vented to limit the external temperature of surfaces not normally in human contact to 185° F

Color correction shall be achieved by a cylindrical type urc H10 glass filter and the dichroic coated reflector. Heat dissipation shall be achieved by absorption into the glass filter and heat transmitted through the dichroic coating on the reflector

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The light shall be supplied with two handles, one permanently attached, and a detachable, sterilizable one. Handle temperature shall not exceed 20°F above ambient

The lighthead shall be supplied with an on/off switch and shall incorporate a "soft start" feature to eliminate initial filament "shock"

The light shall be field repairable with common hand tools, with lamp replacement taking less than one minute and requiring no tools

The surgical field light shall be packed in a rainproof chest which shall be an olive drab color

Finish

Finish shall be highly fungus, mildew, corrosion and salt-spray resistant The unit shall have the manufacturer's highest quality commercially available, easily cleaned, washable finish which incorporates the following paint process and finish

1 Paint process and finish

a The following process shall be minimum requirements for tripod, suspension tube, yoke assembly and arms

(1) Alkaline spray clean
(2) Rinse
(3) Zinc phosphate
(4) Rinse
(5) Non-chromic acid rinse
(6) Polyurethane color coat
(7) Air Dry
(8) Polyurethane texture coat
(9) Bake

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b The following process shall be minimum requirements for lamphead.

(1) Polyurethane color coat
(2) Air dry
(3) Polyurethane texture coat
(4) Bake

Standards and approvals:

Fixture shall meet the requirements of

National Electric Code as prescribed in National Fire Protection Association Pamphlet NFPA No 56A

UL 544 - Medical and Dental equipment
UL 153 - Portable electric lamps
UL 498 - Electrical attachment plugs and receptacles

Workmanship The surgical field light shall be free from defects which detract from its appearance or impair its serviceability.

Unit Each (EA) One light, as specified constitutes one unit

Quality Assurance Provisions

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

Records Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request at any time or from time to time during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate

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Inspection Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item

Sampling for examination Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 2.5 (percent defective) and an inspection level of II The unit of product for examination purposes shall be one light

Sampling for tests Sampling for tests, including dimensional test, shall be conducted in accordance with MIL-STD-105, with an AQL of 2.5 (percent defective) and an inspection level of II The unit of product for test purposes shall be one light

Tests Tests shall be conducted to determine compliance with specification requirements Where feasible, the same sample shall be used for the determination of two or more test characteristics Test shall include but shall not be limited to the following

System test A system operation simulation test shall be performed to insure proper functioning of all applicable components

Electrical conductivity test The light while mounted on the casters, shall be tested for electrical conductivity using the equipment described in NFPA 56A

Metric products Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable

The contracting officer has the option of accepting or rejecting the product

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Contractor certification The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements

Federal Food Drug and Cosmetic Act If the product covered by this document has been determined by the U.S Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances and regulations

Recovered materials The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23 4 to the maximum extent practical

Preservation, packaging, packing, labeling, and marking Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc , involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and or order

Ordering data Intermediate/exterior package quantities, labeling and marking must be specified in the contract and/or order

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NOTES 1 The following National Stock Number is covered by this document

6530-01-316-8733

2 The surgical field light shall be compatible with Medical Instrument and Supply Set, Field No. 3 Chest (NSN 6545-00-914-3480), modified in accordance with DPSC Drawing No 25010

MILITARY INTERESTS

Custodians

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES

VA-OSS
USPHS
FDA-MPQAS

Project Number 6530-2198

Location: LIGHTCID 1BC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-54251	LIGHT SURGICAL FIELD

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
3b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional	b. WORK TELEPHONE NUMBER (Include Area Code) – Optional
c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional	8. DATE OF SUBMISSION (YYMMDD)

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